Introduction: Doped inert cryogenic crystals offer a promising platform for fundamental physics research, including high-precision measurements and searches for physics beyond the standard model. By encasing atoms or molecules (dopants) inside an inert cryogenic crystal, one creates an 'artificial vacuum'. This pristine environment enables the precise measurements of quantum properties such as the electron electric dipole moment (eEDM) among many others [1]. Electron EDM measurements are among the most sensitive probes for new physics which could explain some of the biggest unanswered questions in fundamental physics such as the matter-antimatter asymmetry. Additionally, these doped crystals have potential applications in dark matter searches and in quantum information science for quantum sensors and simulators. In this project, I will study the parameter space to grow these crystals and the systematic effect it has on their structure and measurement capabilities.

Despite their potential, doped crystals currently suffer from what many believe are structural defects limiting their performance [2]. While much research has gone into exploring the use of different dopants and host crystals, and the resulting measurements that can be made, little has been done to fully understand the relationship between crystal growth parameters and resulting crystal structure. Therefore, the goal of this project is to explore this relationship, characterizing how growth rate, dopant density, and beam velocity (beam of dopant and inert gas sent to growth substrate) influence crystal structure and optical clarity. This study will develop insight into the formation and structure of doped cryogenic crystals, which is foundational for all the work listed above.

Background/Lit Review: A big source of interest in doped cryogenic crystals come from their ability to greatly increase statistical sensitivity, as they can isolate a large number of atoms or molecules for simultaneous measurements. For example, one of the most competitive eEDM measurements, by ACME-II, spent 350 hours collecting data in a beam of a few hundred thousand molecules flowing through every second [3]. Doped crystals can contain orders of magnitude more molecules. A doped cryogenic crystal with 10^{13} dopants/mm^3 could produce an eEDM measurement with the same uncertainty as ACME-II in seconds instead of weeks [7].To do this, however, the dopants need a high enough 'ensemble transverse spin relaxation time' otherwise known as T2* or a coherence time (time before a quantum system loses stability from environmental effects). With the above dopant density, T2* must be at least a ms. Previous work has achieved T2* ~ 1 ms with cesium-doped helium crystals, but the pressure required to solidify helium makes growth of crystals with high dopant density technically intractable [4]. Others have grown neon, parahydrogen, and argon crystals doped with alkali-atoms with the desired dopant densities but insufficient T2* times (~10-100us) [2, 5, 6]. Some claim the dominant limitation on coherence time comes from crystal inhomogeneities [2]. Dopants which experience different local geometry inside the same crystal will dephase, decreasing the size of the signal. Hosting dopants inside a perfect single crystal provides uniform local geometry and could increase the coherence time. The Odom lab will explore this relationship and analyze dominant sources of decoherence, growing rubidium (Rb) doped cryogenic argon (Ar) crystals.

Many factors, such as beam velocity, dopant density, growth rate, and growth substrate temperature, all interact and affect the crystal. Beam velocity can influence how dopants incorporate into the lattice structure, and the surface dynamics of incorporation are only qualitatively understood. Little has been done to study how growth rate impacts crystal quality, but most groups have chosen to grow around 0.5-2 μm/min without much explanation [10, 8]. Additionally, optical clarity has been an issue for some crystals hindering the use of optical techniques needed to measure many properties of the crystal and dopant [10]. Although this is

typically attributed to growth substrate temperature, it has not been systematically studied and it is believed other parameters may cause similar issues. In general, a systematic study into the relationship between growth parameters and crystal properties has yet to be done. Better understanding this is an important step towards understanding the growth of doped cryogenic crystals, necessary to fully utilize their theorized capabilities for studying physics beyond the standard model. I have already designed and am assembling a dual-species beam source (Ar and Rb mix together before sent to growth substrate) to grow these crystals, so I will lead the effort to study this beam and study the crystals it produces. By doing so, I will **characterize the parameters set by my dual-species beam–growth rate, beam velocity, and dopant density–and map this to the optical clarity and structure of doped crystals.**

Methodology: This project will be in two phases: first, the characterization of the dual-species beam, and second, the growth and analysis of doped crystals under varying growth conditions. The goal is to establish a systematic relationship between beam parameters and crystal properties: optical clarity and uniformity.

In the first phase, beam velocity will be determined observing what's known as doppler shifts in absorption spectroscopy. The motion of an atomic beam causes a shift in Rb transition frequencies [9]. I will probe this shift with a laser to deduce the velocity of this atomic beam. Growth rate can be determined by a flowmeter measuring the input rate of Ar along with the known system geometry. Dopant density can be determined using absorption spectroscopy, which, with the beam velocity measurements, tells us the flow rate of the Rb atoms. The ratio between this and Ar flow rate is the dopant density [10]. My dual-species beam source was designed for tunability of these parameters, and should allow us to explore growth rates between 0.1 μm/min to 10 μm/min, dopant densities between 10^{11} and 10^{15} dopants/mm^3, and velocities between 250 and 600 m/s. An initial study of this 3-dimensional parameter space will involve studying only three values for each of these parameters (e.g. growth rates of 0.1, 1, and 10 μm/min), one of which will be what other groups have previously used, and all 27 (3^3) combinations that are possible from them. The first phase is primarily aimed at identifying the beam source settings needed to achieve the growth parameters we are interested in. In the second phase, I will use these parameters to grow crystals, while a PhD student will characterize their structure and optical clarity using optical and x-ray techniques. Although I will assist in this analysis, my focus remains on mapping the growth parameter space to the resulting crystal phase space produced through careful characterization of my dual-species beam.

Qualifications: I have been a researcher in the Odom Lab since Fall 2023, where I have designed and began assembling the dual-species beam. I have helped in the design process of our main experiment chamber to grow and characterize crystals, giving me insight into our larger research goals. I also have acquired experience working with ultra high vacuum systems and writing SOPs for laser safety. Prior to this, I worked in the Kovachy Lab, gaining experience in machining and CAD modeling. Having completed my Physics degree requirements by this summer, I will have developed a deep understanding of the underlying physics required for this project, as well as completing two advanced physics labs gives me the data analysis and critical thinking skills necessary to this task. Additionally, I work closely with the PhD student and Professor Odom in our lab, attending and presenting my work in weekly meetings. I also communicate with PhD students from Kovachy lab and other students in our research center as they have experience in many of the techniques used in this project. Furthermore, I hope to use the findings of this project as a contribution to a senior thesis and help build a foundation in my research experience and skills to support my ultimate goal of obtaining a PhD in Physics.

References:

Note: Citations in APS style

[1] A. C. Vutha, M. Horbatsch, and E. A. Hessels. Oriented polar molecules in a solid inert-gas matrix: a proposed method for measuring the electric dipole moment of the electron. *Atoms*, 6(1), 3 (2018).

[2] S. Upadhyay, U. Dargyte, V. D. Dergachev, R. P. Prater, S. A. Varganov, T. V. Tscherbul, D. Patterson, and J. D. Weinstein. Spin coherence and optical properties of alkali-metal atoms in solid parahydrogen. *Physical Review A*, 100(6), 063419 (2019).

[3] ACME. Improved limit on the electric dipole moment of the electron. *Nature*, 562(7727), 355–360 (2018).

[4] S. I. Kanorsky, S. Lang, S. Lucke, S. B. Ross, T. W. Hänsch, and A. Weis. Millihertz magnetic resonance spectroscopy of Cs atoms in body-centered-cubic ^{4}He. *Physical Review A*, 54, R1010–R1013 (1996).

[5] D. M. Lancaster, U. Dargyte, S. Upadhyay, and J. D. Weinstein. Radiative properties of rubidium atoms trapped in solid neon and parahydrogen. *Physical Review A*, 106, 042810 (2022).

[6] U. Dargyte, D. M. Lancaster, and J. D. Weinstein. Optical and spin-coherence properties of rubidium atoms trapped in solid neon. *Physical Review A*, 104(3), 032611 (2021).

[7] B. Odom. PM: Doped inert single crystals for probing beyond the standard model. *National Science Foundation*, (2022).

[8] U. Dargyte, D. M. Lancaster, and J. D. Weinstein. *Optical and Spin-Coherence Properties of Rubidium Atoms Trapped in Solid Neon*. Manuscript in preparation (2021).

[9] D. O. Sabulsky. *A Cesium-133 Effusive Oven for Ultracold Atomic Experiments*. Bachelor's thesis, The University of Chicago (2014).

[10] S. K. Regmi. *Optical Pumping of Rubidium Spin in a Solid Argon Matri*x. PhD thesis, University of Nevada, Reno (2013).